OPTIMALIZAČNÉ METÓDY

1D

Gradient descent

$$x_{k+1} = x_k - \alpha f'(x_k)$$

$f(x) = x^4 - 2x^2$ $\quad \alpha = 0.05$ $\quad x_0 = 2$

$f' = 4x^3 - 4x$

1. ITERÁCIA:

$x_1 = x_0 - \alpha \cdot f'(x_0)$

$x_1 = 2 - 0.05\,(4 \cdot 2^3 - 4 \cdot 2) = 2 - 0.05\,(24) = 0.6$

2 ITERÁCIA:

$x_2 = x_1 - \alpha f'(x_1)$

$x_2 = 0.6 - 0.05\,(4 \cdot 0.6^3 - 4 \cdot 0.6) = 0.6768$

3. ITERÁCIA:

$x_3 = x_2 - \alpha f'(x_2)$

NEWTONOVA METÓDA

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$f(x) = x^4 - 2x^2$ $\quad x_0 = 2$

$f' = 4x^3 - 4x$

$f'' = 12x^2 - 4$

1. ITERÁCIA

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)} = 2 - \frac{4 \cdot 2^3 - 4 \cdot 2}{12 \cdot 2^2 - 4} = 2 - \frac{28}{44} = 1.3636$$

2 ITERÁCIA

$$x_2 = x_1 - \frac{f'(x_1)}{f''(x_1)} = 1.3636 - \frac{4\,1.3636^3 - 4 \cdot 1.3636}{12 \cdot 1.3636^2 - 4} = 1.1072$$

3 ITERÁCIA

$$x_3 = x_2 - \frac{f'(x_2)}{f''(x_2)} =$$

2D

Newtonova metóda $f(x,y) = x^2 + 2y^2$ $x_0 = [2, 2]$

$$x_{k+1} = x_k - H^{-1} \cdot \nabla f(x_k)$$
$$y_{k+1} = y_k - H^{-1} \cdot \nabla f(y_k)$$

$H = \begin{vmatrix} 2 & 0 \\ 0 & 4 \end{vmatrix}$ $H^{-1} = \begin{vmatrix} 0.5 & 0 \\ 0 & 0.25 \end{vmatrix}$

1 ITERÁCIA: $\nabla f = (2x, 4y)$

$f'_x = 2x$

$f'_y = 4y$

$x_1 = x_0 - H^{-1} \cdot \nabla f(x_0) = 2 - 0.5 \cdot (2\ 2) = 2 - 0.5 \cdot 4 = 0$

$y_1 = y_0 - H^{-1} \nabla f(y_0) = 2 - 0.25\ (4 \cdot 2) = 2 - 0.25\ 8 = 0$

$[0, 0]$

2 ITERÁCIA:

$x_2 = x_1 - H^{-1} \nabla f(x_1)$

$y_2 = y_1 - H^{-1} \nabla f(y_1)$

Postup:

1. Usporiadaj body podľa hodnoty f: $f(A) \le f(B) \le f(C)$.
2. Vypočítaj ťažisko:

best ← great

$$x_0 = \frac{A + B}{2}$$

→ môže sa označovať aj M.

3. **Reflexia:**

$$x_r = x_0 + \alpha(x_0 - C), \quad \alpha = 1 \rightarrow R = M + 1 \cdot (M - W)$$

najhorší bod

$\alpha = 1$

Ak $f(A) \le f(x_r) < f(B)$, nahraď $C = x_r$.

4. **Expanzia** (ak $f(x_r) < f(A)$):

$$x_e = x_0 + \gamma(x_r - x_0), \quad \gamma = 2 \rightarrow E = M + 2 \cdot (R - M)$$

p_m

$\gamma = 2$

Ak $f(x_e) < f(x_r)$, potom $C = x_e$, inak $C = x_r$.

5. **Kontrakcia** (ak $f(x_r) \ge f(B)$):

$$x_c = x_0 + \rho(C - x_0), \quad \rho = 0.5 \rightarrow K = M + \frac{1}{2}(W - M)$$

$\beta = 0.5$

Ak $f(x_c) < f(C)$, nahraď $C = x_c$, inak všetky body zmenši k A:

$$B = A + \sigma(B - A), \quad C = A + \sigma(C - A), \quad \sigma = 0.5$$

Iterácia 1–2: Urob kroky 1–5, aktualizuj body a opakuj, kým zmeny $f(x)$ sú malé.

NELDER- MEAD $f(x,y) = x^2 + 2y^2$ $A[2,2]$ $B[2,1]$ $C[1,2]$

Ohodnotenie vrcholov : $f(A) = 2^2 + 2 \cdot 2^2 = 12$

$f(B) = 2^2 + 2\,1^2 = 6$

$f(C) = 1^2 + 2\,2^2 = 9$

$6 \leq 9 \leq 12$

$B \leq C \leq A$

B ↓ best, C ↓ great, A ↘ worst

1 ITERÁCIA

1) Ťažisko (centroid)

$$x_0 = \frac{B+C}{2} = \frac{[2,1] + [1,2]}{2} = \left[\frac{3}{2}, \frac{3}{2}\right]$$

2) Reflexia

x_h → najhorší bod

$$x_r = x_0 + \alpha(x_0 - A)$$

$$x_r = \left[\frac{3}{2}, \frac{3}{2}\right] + 1\left(\left[\frac{3}{2}, \frac{3}{2}\right] - [2,2]\right) = [1,1]$$

$f(B) \leq f(x_r) < f(C)$ $\quad f(x_r) = 1^2 + 2\,1^2 = 3$

$6 \leq 3 < 9$ ✗

3) $3 < 6 \Rightarrow$ Expanzia

$$x_e = x_0 + \gamma(x_r - x_0)$$

$$x_e = \left[\frac{3}{2}, \frac{3}{2}\right] + 2\left([1,1] - \left[\frac{3}{2}, \frac{3}{2}\right]\right) = [0.5, 0.5]$$

$f(x_e) = 0.5^2 + 2 \cdot 0.5^2 = 0.75$

$f(x_e) < f(x_r)$

$0.75 < 3$

$A \Rightarrow [0.5, 0.5]$

$A[0.5, 0.5]$ $B[2,1]$ $C[1,2]$

2 ITERÁCIA

$f(A) = 0.75$

$f(B) = 6$

$f(C) = 9$

$A \leq B \leq C$

A ↓ best, C ↘ worst

1) Ťažisko :

$$x_0 = \frac{A+B}{2} = \frac{[0.5, 0.5] + [2,1]}{2} = [1.25, 0.75]$$

2) Reflexia :

$$x_r = x_0 + \alpha(x_0 - C) = [1.25, 0.75] + 1([1.25, 0.75] - [1,2]) = [1.5, -0.5]$$

$f(x_r) = 2.75$

$f(A) \leq f(x_r) < f(B)$

$0.75 \leq 2.75 < 6$ ✓ $\quad C \Rightarrow x_r \Rightarrow [1.5, -0.5]$

$A[0.5, 0.5]$ $B[2,1]$ $C[1.5, -0.5]$



3 ITERÀCIA